

Tempotron learning in deep neural networks with direct feedback alignment

Abstract

Limitations in deep spiking learning models hinder our understanding of how biological systems 1) build spike-based representations, 2) solve the credit assignment problem, and 3) learn efficiently without backpropagation (BP). We address this challenge by developing a biologically-plausible deep learning model (TeDFA) that combines spiking neurons with local Tempotron learning and global Direct Feedback Alignment. This mimics three-factor learning where DFA models dopaminergic feedback pathways. We find that TeDFA is competitive with BP-trained spiking neural networks (SNNs) and MLPs on various supervised classification tasks, on both static and dynamic inputs. Additionally, a reinforcement learning (RL) variant using biological temporal difference error (TeDFA- δ) excels at online control tasks such as the cartpole problem where it outperforms MLPs trained with BP. We find that this improvement in RL stems from temporal integration of the history of environment states in spiking neurons. Although TeDFA can be implemented with multiple spikes per neuron, here we have constrained all hidden layer neurons to emit at most one spike, rendering it highly compute efficient and showing that spike timing alone is sufficient for complex tasks in deep networks. Interestingly, BP-trained SNNs perform relatively poorly when constrained to a single spike. The model develops learned structured spatiotemporal latent representations of class memberships (supervised classification tasks) and joint policy and state variables (RL). Optimal performance is achieved for intermediate membrane time constants (except for the relatively simple MNIST task) and smoothly rising PSP temporal kernels.

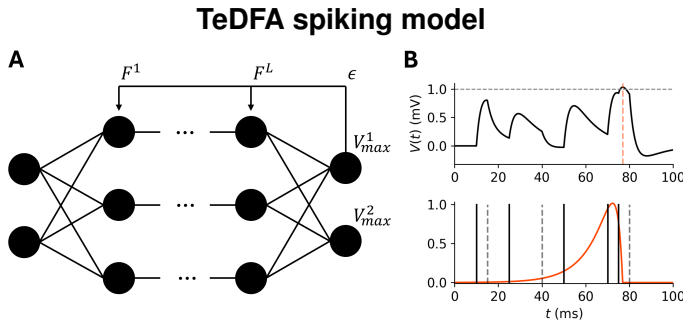


Figure 1: **A** The TeDFA model. The output neurons directly provide feedback to hidden layer neurons via the TeDFA learning rule. **B** A single Tempotron neuron with a sample trace of membrane potential over time (top) and the corresponding Tempotron learning rule with $\phi_{ij}(t_{\max}^j - t_i^j)$ (bottom, orange).

We use a network consisting of Leaky Integrate-and-Fire spiking neurons, where each neuron's membrane potential $V(t) = \sum_i \omega_i \phi_i(t - t_i)$ integrates postsynaptic potentials (PSPs) from incoming spikes $\phi_i(t) = \sum_{t_i} (e^{-(t-t_i)/\tau} - e^{-(t-t_i)/\tau_s})$ with time constants τ and τ_s , and t_i incoming spike times at synapse i . A spike is emitted when $V(t) \geq 1$, which resets and fixes the

membrane potential to zero for the rest of the simulation. We use the local Tempotron rule [1] and extend this to deep networks using Direct Feedback Alignment (DFA) [2,3] through the proposed TeDFA rule, where weight updates combine global error signals e_k^L with local temporal activity ϕ using fixed random feedback matrices F_{jk}^l (Fig. 1). The update rule is $\Delta \omega_{ji}^l \propto -(F_{jk}^l \cdot e_k^L) \odot \phi_{ij}(t_{\max}^j - t_i^j)$.

Results

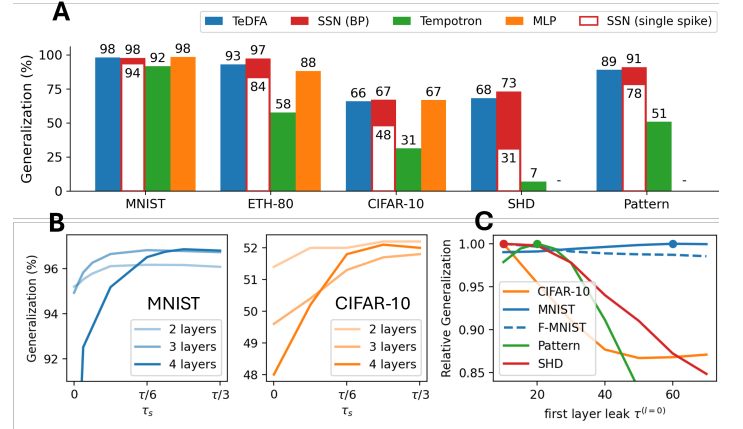


Figure 2: **A** TeDFA performs similar to MLP on various static classification tasks and outperforms BP-trained SNN on both static tasks and spiking tasks. “SHD” is the Spiking Heidelberg Digits task. “Pattern” is a task where spike patterns occur a random number of times on top of background noise; the pattern with most occurrences is the correct label. All models across tasks used a simulation window of 100 timesteps and 2 hidden layers of 512 neurons each, except for CIFAR-10 with 6 hidden layers. Spiking models encode image pixels with time-to-first-spike. **B** A smooth two-sided PSP kernel helps to generalize, especially with network depth. **C** On complex tasks and the pattern task with a determined information window, choosing sufficiently small PSP kernels is necessary for learning. Generalization is the averaged test accuracy.

Supervised learning We test TeDFA on various supervised classification tasks (Fig. 2A). Across tasks, TeDFA’s performance is similar to MLPs and SNNs trained with BP, while consistently outperforming the single layer Tempotron baseline and the single-spike SNN. The model performs slightly worse than an MLP, except on ETH-80 where the spike encoding potentially improves learning of rotation invariant features, based on an ablation study with rotated CIFAR-10. On CIFAR-10, a deeper network (6 hidden layers) and data augmentation were used, which shows that TeDFA benefits from depth. We also tested on two sequential classification tasks (Pattern and SHD), where the single-layer Tempotron and BP-SNN with one spike output fail to generalize well. For all tasks, and especially for deeper architectures, it was important that τ_s is sufficiently large to ensure that the PSP kernel is smooth (Fig. 2B). For complex tasks (CIFAR-10 and the spiking

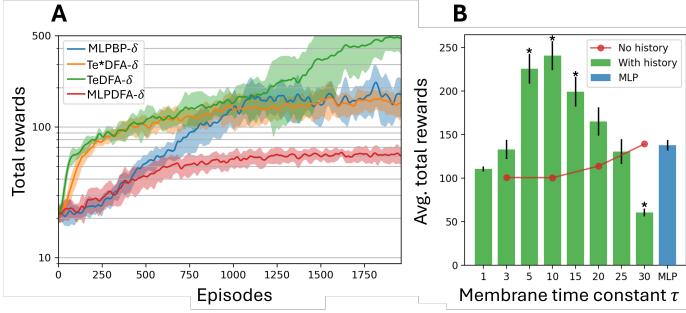


Figure 3: **A** The Tempotron model learns faster and outperforms MLPs on the online cartpole task, where the objective is to balance a pole on a moving cart. Inputs states were encoded as a constant current for TeDFA. Shown are the total rewards averaged over 20 runs with standard deviations. TeDFA- δ is the full model with history, Te*DFA- δ is without state history. **B** TeDFA- δ total rewards averaged over 10 runs (after 1000 episodes) on cartpole for different values of τ , with and without history. Simulation time per environment step is $T = 30$.

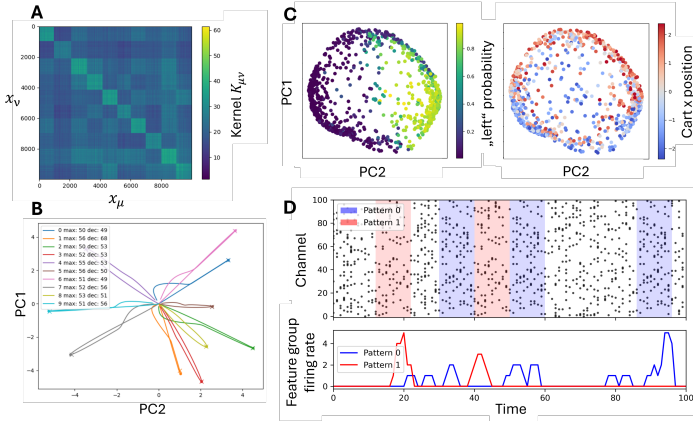


Figure 4: The TeDFA model learns useful spatio-temporal representations. **A** MNIST: Feature similarity kernel K between pairs of inputs. **B** MNIST: Average feature trajectories $\xi(t)$ for each class. **C** RL cartpole: Integrated feature trajectories ξ for 100 sample inputs after training, color-coded by policy (left) and the x position of the cart (right). The model has to learn to balance the stick and evade the x -borders with the same action. **D** Pattern task: the feature layer organizes in two groups, each of which responds to one of the two patterns present in the task.

tasks) it was necessary that the PSP kernel widens with depth for temporal feature integration (Fig. 2C), similar to the brain [4].

Reinforcement learning For RL settings, we extend TeDFA to TeDFA- δ using the Temporal Difference (TD) error and a deep, shared hidden-layers actor-critic setup. In particular, the DFA feedback is now $F e_t = F^\pi e_t^\pi + F^v e_t^v = F^\pi (\tilde{g}_t \cdot \delta_t) + F^v \delta_t$, where $\tilde{g}_t$ is a normalized one-hot action vector, and F^π, F^v are fixed random feedback matrices for actor and critic pathways respectively, similar to dopaminergic pathways in the brain. We evaluate TeDFA- δ on the cartpole task, where small policy changes can lead to failure and online learning is difficult due to long trial durations. TeDFA- δ achieves near-perfect performance, outperforming both MLP with BP and a memory-less variant, Te*DFA- δ , which resets membrane potential after each step (Fig. 3A). The 10x

faster convergence of both spiking models suggests temporal integration is crucial - confirmed by the performance drop when using larger time constants ($\tau \sim T$; Fig. 3B). We also tested on the online acrobat task and a continual 10-armed bandit task, where TeDFA- δ learns faster and more reliable than MLPs with BP.

Learned Representations To visualize time-dependent features learned by TeDFA, we calculate a global autocorrelation matrix for each input x via the correlation matrix of the $\phi_x(t)$ over synapses i : $C_{ii'} = \frac{1}{T} \sum_x \int dt \phi_i(t - t_i^x) \phi_{i'}^T(t - t_i^x)$. We project $\phi_x(t)$ onto the first principal components of C to get feature trajectories $\xi(t)$. Additionally, we can calculate the feature similarity kernel between two inputs μ and ν , namely $K_{\mu\nu}$, by averaging over synapses instead of inputs.

On classification tasks, we see the typical feature similarity structure between inputs of the same class (Fig. 4A). As the features evolve in time in spiking networks we can plot the class averaged $\xi^{\text{class}}(t)$ (Fig. 4B). These features are clearly separable and have a maximal time point of explained global spatial autocorrelation which coincides with the decision time of the output layer across tasks. Separability of $\xi^{\text{class}}(t)$ grows both with depth and training time (not shown here). For more complex tasks like CIFAR-10 the class trajectories ξ are generally less separable. In RL, TeDFA- δ develops structured representations of learned policies in feature space. The model learns a circular representations of cartpole states where policy and state are jointly encoded (Fig. 4C) and prepares future actions through membrane potential history. For the Pattern task, feature layer neurons split into groups, each selective to one of the patterns in the task. The decision is then made by comparing the spike counts from those groups (Fig. 4D).

Conclusion

We present TeDFA, a biologically plausible deep spiking model that, despite DFA's constraints, is competitive with MLPs and SNNs trained with BP on static and sequential supervised tasks, and outperforms SNNs with single-spikes. On online RL tasks, TeDFA outperforms MLPs with BP through temporal integration. This suggests biological networks may compensate for weaker global signals and imperfect credit assignment with local learning, temporal dynamics and robust representations. Future work will explore scalability, neural tasks, and assess biological alignment of learned representations [5].

References [1] Güttig, R., & Sompolinsky, H. (2006). *Nat Neurosci*, 9, 420-428. [2] Nøkland, A. (2016). *Arxiv*, 1609.01596 [3] Shi, C., Wang, T., He, J., Zhang, J., Liu, L., & Wu, N. (2021). *IEEE Transactions on Circuits and Systems II*, 68(5), 1581-158 [4] Gjorgjieva, J., Drion, G., & Marder, E. (2016). *Current Opinion in Neurobiology*, 37, 44-52. [5] Kumar, M. G., Manooogian, A., Qian, B., Pehlevan, C., & Rhoads, S. A. (2025). *bioRxiv*, 202503.